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Editor-in-Chief

The American Mathematical Monthly
Mathematical Association of America

Dear Editor,

I submit for your consideration the manuscript:

**When Jensen's Equation Has a One-Line Proof:
Three Regularity Hypotheses That Become Unnecessary**

for publication in *The American Mathematical Monthly* as an **Article**.

What the paper does. The paper shows that the continuous-coefficient Jensen equation $G(pu_1 + (1-p)u_2) = pG(u_1) + (1-p)G(u_2)$ for all $u_1, u_2 \in [0, M]$ and all $p \in [0, 1]$ forces G to be affine with *no regularity hypothesis*: one substitution ($u_1 := M$, $u_2 := 0$, and noting that p is already free) gives the explicit formula in two lines. The paper accompanies this one-line observation with (i) a six-row dictionary comparing it with the well-known case of the discrete-coefficient equation (where Hamel's pathology is real and regularity *is* needed), (ii) a structural explanation of why the two cases differ, and (iii) a survey of application areas—utility theory, information-theoretic entropy, and surrogate-loss calibration—where the same vestigial hypotheses recur.

Why the Monthly. The result is elementary enough for a second-year analysis student to verify in an afternoon, yet it is consistently missed in practice: the functional equations literature treats the two forms of Jensen's equation in adjacent sections without flagging the asymmetry, and modern applied work (utility theory, calibration, entropy) carries defensive regularity assumptions that are unnecessary precisely in the continuous-coefficient case. The Monthly is the natural home for an expository note that makes this asymmetry explicit, provides the dictionary, and invites readers to audit their own work.

Novelty and relationship to prior art. The one-line closure itself is not new: it appears in compressed form in Aczél (1966) and Kuczma (2009), and no claim to its discovery is made here. The contribution is the *explicit packaging*: the dictionary contrasting the two Jensen forms, the structural explanation of the Hamel-pathology mechanism, and the identification of the recurrence pattern across application domains. The appendix describes how the observation surfaced through a proof-assistant formalization exercise, a provenance that the author believes adds interest for Monthly readers.

Submission details.

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- **Double-blind:** Yes. The anonymous PDF contains no author-identifying information. Self-citations to a companion manuscript appear as “[citation removed for review]”. A separate title page is attached.
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- **Competing interests:** None.
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I look forward to the referees’ comments.

Sincerely,

[Author Name]